PROBLEMS SHEET 1 • OFFICIAL SOLUTIONS

Department of Mathematics • UC3M Aerospace Engineering

Chapter 1 Problem Set: Introduction, Modeling and Classification

Comprehensive, rigorous analytical solutions to all 10 problems from `ProblemsCh1.pdf`, developed step-by-step with zero algebraic jumps and structured according to our 4-Phase Engineering Framework.

Ground Truth: ProblemsCh1.pdf • Official Department Problem Sheet

OFFICIAL MATERIAL ProblemsCh1.pdf Robinson (2004).pdf

PROBLEM 1.1 Taxonomy • Order • Linearity

Classification of Ten Fundamental Differential Equations

Original statement · ProblemsCh1.pdf

PHASE 1: PROBLEM STATEMENT & HYPOTHESES

Problem statement: Classify the following equations as ordinary or partial; give their order; and state whether they are linear or nonlinear. In each case identify the dependent and independent variables:

1. (i) **Bessel's equation** (ν is a parameter): $x^2y'' + xy' + (x^2 - \nu^2)y = 0$
2. (ii) **Burgers' equation** (ν is a parameter): $\frac{\partial u}{\partial t} - \nu\frac{\partial^2 u}{\partial x^2} + u\frac{\partial u}{\partial x} = 0$
3. (iii) **Duffing's equation** (m, k, a, b are parameters): $m\ddot{x} + kx = ax - bx^3$
4. (iv) $\frac{dy}{dt} = t - y^2$
5. (v) **The wave equation** (c is a parameter): $\frac{\partial^2 y}{\partial t^2} = c^2\frac{\partial^2 y}{\partial x^2}$
6. (vi) **Newton's law of cooling** (k is a parameter and $T_A(t)$ is a specified function): $\frac{dT}{dt} = -k(T - T_A(t))$
7. (vii) **The logistic population model** (k is a parameter): $\frac{dp}{dt} = kp(1 - p)$
8. (viii) **Newton's second law** for a particle of mass m moving in a potential $V(x)$: $m\ddot{x} = -V'(x)$
9. (ix) **The coupled equations:** $\dot{x} = x(4 - 2x - y), \quad \dot{y} = y(9 - 3x - 3y)$
10. (x) $\frac{d\mathbf{x}}{dt} = A\mathbf{x}$, where $\mathbf{x}$ is an n-component vector and A is an $n \times n$ matrix.

PHASE 2: FUNDAMENTAL FORMULATION & CRITERIA

We apply the four foundational mathematical tests established by the UC3M Department of Mathematics:

- **Type (ODE vs PDE):** An ODE depends on exactly 1 independent variable; a PDE depends on 2 or more independent variables.
- **Order:** The highest derivative index present in the equation.
- **Dependent vs Independent:** Independent is the differentiation variable(s); dependent is the unknown state function(s).
- **Linearity (Cátedra Standard):** An equation is linear if and only if it is of degree 1 in the dependent variable and all its derivatives, with no mutual products or transcendental compositions: $\sum_{j=0}^{n} a_j(x)y^{(j)} = b(x)$.

PHASE 3: STEP-BY-STEP MATHEMATICAL DERIVATION

(i) Bessel's equation: $x^2y'' + xy' + (x^2 - \nu^2)y = 0$
Independent variable: x (single $\implies$ **ODE**). Dependent variable: $y(x)$.
Derivatives present: y' and y''. Highest derivative is order 2 $\implies$ **Order 2**.
Linearity: Matches $a_2(x)y'' + a_1(x)y' + a_0(x)y = b(x)$ with $a_2(x) = x^2, a_1(x) = x, a_0(x) = x^2 - \nu^2, b(x) = 0$. All y, y', y'' appear linearly $\implies$ **Linear**.

(ii) Burgers' equation: $\frac{\partial u}{\partial t} - \nu\frac{\partial^2 u}{\partial x^2} + u\frac{\partial u}{\partial x} = 0$
Independent variables: t, x (two $\implies$ **PDE**). Dependent variable: $u(x, t)$.
Highest derivative: $\frac{\partial^2 u}{\partial x^2}$ $\implies$ **Order 2**.

Linearity: The convective acceleration term $u\frac{\partial u}{\partial x}$ is a product of the dependent variable and its derivative $\implies$ **Nonlinear**.

(iii) Duffing's equation: $m\ddot{x} + kx = ax - bx^3$
Overdot signifies time derivative: $\ddot{x} = \frac{d^2x}{dt^2}$. Independent variable: t $\implies$ **ODE**. Dependent variable: $x(t)$.
Highest derivative: $\ddot{x}$ $\implies$ **Order 2**.
Linearity: Rearranging gives $m\ddot{x} + (k-a)x + bx^3 = 0$. If $b \neq 0$, the cubic power x^3 violates linearity $\implies$ **Nonlinear**.

(iv) Riccati Type: $\frac{dy}{dt} = t - y^2$
Independent variable: t $\implies$ **ODE**. Dependent variable: $y(t)$. Highest derivative: $\frac{dy}{dt}$ $\implies$ **Order 1**.
Linearity: The term y^2 is quadratic in the dependent variable $\implies$ **Nonlinear**.

(v) Wave Equation: $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$
Independent variables: t, x $\implies$ **PDE**. Dependent variable: $y(x,t)$ (displacement).
Highest derivative: $\frac{\partial^2 y}{\partial t^2}$, $\frac{\partial^2 y}{\partial x^2}$ $\implies$ **Order 2**.
Linearity: $\frac{\partial^2 y}{\partial t^2} - c^2 \frac{\partial^2 y}{\partial x^2} = 0$ is a homogeneous linear differential operator $\implies$ **Linear**.

(vi) Newton's Law of Cooling: $\frac{dT}{dt} = -k(T - T_A(t))$
Independent variable: t $\implies$ **ODE**. Dependent variable: $T(t)$. Highest derivative: $\frac{dT}{dt}$ $\implies$ **Order 1**.
Linearity: Rearranges to $\frac{dT}{dt} + kT = kT_A(t)$, matching $a_1(t)\dot{T} + a_0(t)T = b(t)$ $\implies$ **Linear**.

(vii) Logistic Model: $\frac{dp}{dt} = kp(1-p)$
Independent variable: t $\implies$ **ODE**. Dependent variable: $p(t)$. Highest derivative: $\frac{dp}{dt}$ $\implies$ **Order 1**.
Linearity: Expanding yields $\frac{dp}{dt} - kp + kp^2 = 0$. The term kp^2 is nonlinear in p $\implies$ **Nonlinear**.

(viii) Particle in Potential: $m\ddot{x} = -V'(x)$
Independent variable: t $\implies$ **ODE**. Dependent variable: $x(t)$. Highest derivative: $\ddot{x}$ $\implies$ **Order 2**.
Linearity: The force $-V'(x)$ is in general a nonlinear function of x (linear only if $V(x) = \frac{1}{2}\kappa x^2$) $\implies$ **Nonlinear**.

(ix) Coupled Competing Species: $\dot{x} = x(4 - 2x - y), \quad \dot{y} = y(9 - 3x - 3y)$
Independent variable: t $\implies$ **System of ODEs**. Dependent variables: $x(t), y(t)$. Highest derivative: 1st order $\implies$ **Order 1**.
Linearity: Expanding gives $\dot{x} = 4x - 2x^2 - xy$ and $\dot{y} = 9y - 3xy - 3y^2$, containing quadratic terms (x^2, y^2) and cross product (xy) $\implies$ **Nonlinear**.

(x) Vector Matrix System: $\frac{d\mathbf{x}}{dt} = A\mathbf{x}$
Independent variable: t $\implies$ **System of ODEs**. Dependent variables: $\mathbf{x}(t) \in \mathbb{R}^n$. Highest derivative: 1st order $\implies$ **Order 1**.
Linearity: Matrix multiplication is a linear operator: $A(c_1\mathbf{x}_1 + c_2\mathbf{x}_2) = c_1A\mathbf{x}_1 + c_2A\mathbf{x}_2$ $\implies$ **Linear**.

PHASE 4: FINAL RESULT & ANALYSIS

MASTER CLASSIFICATION MATRIX

Item	Equation	Type	Order	Linearity	Dependent	Independent
(i)	$x^2y'' + xy' + (x^2 - \nu^2)y = 0$	ODE	2	Linear	y	x
(ii)	$u_t - \nu u_{xx} + uu_x = 0$	PDE	2	Nonlinear	u	t, x
(iii)	$m\ddot{x} + kx = ax - bx^3$	ODE	2	Nonlinear	x	t
(iv)	$y' = t - y^2$	ODE	1	Nonlinear	y	t
(v)	$y_{tt} = c^2 y_{xx}$	PDE	2	Linear	y	t, x
(vi)	$T' = -k(T - T_A(t))$	ODE	1	Linear	T	t
(vii)	$p' = kp(1 - p)$	ODE	1	Nonlinear	p	t
(viii)	$m\ddot{x} = -V'(x)$	ODE	2	Nonlinear	x	t
(ix)	$\dot{x} = x(4 - 2x - y), \dot{y} = y(9 - 3x - 3y)$	ODE System	1	Nonlinear	x, y	t
(x)	$\dot{\mathbf{x}} = A\mathbf{x}$	ODE System	1	Linear	$\mathbf{x}$	t

PROBLEM 1.2

Separable ODE • Asymptotics • Doubling Time

Malthusian Population Dynamics & Doubling Time

Original statement · ProblemsCh1.pdf

PHASE 1: PROBLEM STATEMENT & HYPOTHESES

Problem statement: Let $x(t)$ be the population of a certain species as a function of time, and suppose that the rate of change of this population is proportional to the existing population at each time, with constant of proportionality $k \in \mathbb{R}$.

(i) Write a differential equation for $x(t)$.
(ii) Solve it, determine the behavior of $x(t)$ as $t \to \infty$ and interpret it depending on the sign of k.
(iii) If the initial population is $x(0) = x_0$ and $k > 0$, determine the time needed for the population to double.

PHASE 2: FUNDAMENTAL FORMULATION

The rate of change is the time derivative $\frac{dx}{dt}$. Proportionality to $x(t)$ with constant k yields:

$$\frac{dx}{dt} = kx(t), \quad x(0) = x_0 > 0$$

This is a first-order linear separable homogeneous ODE.

PHASE 3: STEP-BY-STEP DERIVATION

Part (i): By definition, the ODE is:

$$\mathbf{\frac{dx}{dt} = kx} \tag{1.2.1}$$

Part (ii): Separate variables assuming $x > 0$:

$$\frac{1}{x}\,dx = k\,dt$$

Integrate both sides from initial state $(0, x_0)$ to arbitrary state $(t, x(t))$:

$$\int_{x_0}^{x(t)} \frac{1}{\xi}\,d\xi = \int_0^t k\,ds \implies [\ln \xi]_{x_0}^{x(t)} = k[s]_0^t \implies \ln\left(\frac{x(t)}{x_0}\right) = kt$$

Exponentiate both sides:

$$\frac{x(t)}{x_0} = e^{kt} \implies \mathbf{x(t) = x_0 e^{kt}} \tag{1.2.2}$$

Asymptotic limits as $t \to \infty$:

- If $k > 0$: $\lim_{t\to\infty} x(t) = \lim_{t\to\infty} x_0 e^{kt} = +\infty$ (Exponential Malthusian explosion; natality exceeds mortality).
- If $k = 0$: $x(t) = x_0 e^0 = x_0$ (Static equilibrium; births perfectly balance deaths).
- If $k < 0$: $\lim_{t\to\infty} x(t) = \lim_{t\to\infty} x_0 e^{-|k|t} = 0$ (Exponential extinction; mortality exceeds natality).

Part (iii): Doubling Time t_d:

Set $x(t_d) = 2x_0$ with $k > 0$:

$$x_0 e^{kt_d} = 2x_0 \implies e^{kt_d} = 2$$

Take the natural logarithm:

$$kt_d = \ln 2 \implies \mathbf{t_d = \frac{\ln 2}{k}} \tag{1.2.3}$$

PHASE 4: FINAL RESULT & DIMENSIONAL VERIFICATION

BOXED RESULT

$$x(t) = x_0 e^{kt}, \quad t_d = \frac{\ln 2}{k}$$

Dimensional Verification: $[k] = \text{time}^{-1} \implies [t_d] = \frac{1}{[k]} = \text{time}$, which is dimensionally homogeneous.

PROBLEM 1.3

Half-Life • Nuclear Decay • Space RTG

Plutonium-239 Radioactive Decay & Decay Constant

Original statement · ProblemsCh1.pdf

PHASE 1: PROBLEM STATEMENT & HYPOTHESES

Problem statement: Plutonium 239, virtually nonexistent in nature, is one of the radioactive materials used in the production of nuclear weapons, and is a byproduct of the generation of power in a nuclear reactor. Its half-life (i.e., the time it takes for half of the radioactive isotope to decay) is approximately 24,000 years.

Using $\frac{dN}{dt} = -kN$ as an appropriate model for radioactive decay, where $N = N(t)$ is the number of isotopes at time t, what is the value of k that should be used for this isotope?

PHASE 2: FUNDAMENTAL FORMULATION

Radioactive disintegration follows the first-order linear ODE $\frac{dN}{dt} = -kN$ ($k > 0$). By definition of half-life $t_{1/2} = 24{,}000$ yr, the isotope quantity at $t = t_{1/2}$ satisfies $N(t_{1/2}) = \frac{1}{2}N_0$.

PHASE 3: STEP-BY-STEP DERIVATION

Separate variables and integrate from $t = 0$ ($N = N_0$) to t:

$$\int_{N_0}^{N(t)} \frac{d\xi}{\xi} = \int_0^t -k\, ds \implies \ln\left(\frac{N(t)}{N_0}\right) = -kt \implies N(t) = N_0 e^{-kt}$$

Substitute $t = t_{1/2}$ and $N(t_{1/2}) = \frac{1}{2}N_0$:

$$N_0 e^{-kt_{1/2}} = \frac{1}{2}N_0 \implies e^{-kt_{1/2}} = 2^{-1}$$

Apply the natural logarithm to both sides:

$$\ln\left(e^{-kt_{1/2}}\right) = \ln\left(2^{-1}\right) \implies -kt_{1/2} = -\ln 2 \implies \mathbf{k = \frac{\ln 2}{t_{1/2}}}$$

Numerical evaluation with $t_{1/2} = 24{,}000$ years and $\ln 2 \approx 0.69314718$:

$$k = \frac{0.69314718}{24{,}000}\ \text{yr}^{-1} \approx \mathbf{2.8881 \times 10^{-5}}\ \text{yr}^{-1}$$

In SI units ($1\ \text{yr} \approx 365.25 \times 86400\ \text{s} = 3.15576 \times 10^{7}\ \text{s}$):

$$k = \frac{2.8881 \times 10^{-5}}{3.15576 \times 10^{7}}\ \text{s}^{-1} \approx \mathbf{9.1519 \times 10^{-13}}\ \text{s}^{-1}$$

PHASE 4: FINAL RESULT & ANALYSIS

BOXED RESULT

$$k = \frac{\ln 2}{24{,}000} \approx 2.8881 \times 10^{-5}\ \text{yr}^{-1} \approx 9.1519 \times 10^{-13}\ \text{s}^{-1}$$

Physical Interpretation: The extremely minute decay constant means that Plutonium-239 remains radioactive, essentially at constant activity, over many centuries (after $10\,t_{1/2} = 240{,}000$ years its activity is still $(1/2)^{10} \approx 0.098\%$ of the initial value). Note that space-probe RTGs (Voyager, Cassini, Curiosity) use the shorter-lived ^{238}Pu ($t_{1/2} \approx 87.7$ years), not ^{239}Pu.

PROBLEM 1.4

Lumped Thermal Model • Variable Identification

Newton's Law of Cooling Modeling

Original statement · ProblemsCh1.pdf, p. 2

PHASE 1: PROBLEM STATEMENT & HYPOTHESES

> **Problem statement:** A cup of hot chocolate is initially at temperature $T(0)$ and is left in the room with an ambient temperature $T_A(t)$. The rate of cooling of the hot chocolate is proportional to the difference between the current temperature $T(t)$ and the ambient temperature.
>
> (i) Which are the dependent variable, the independent variable, and the parameters?
> (ii) Write a linear differential equation for the temperature of the hot chocolate that takes into account the heat exchange with the room. Such differential equation is known as Newton's law of cooling.

PHASE 2: FUNDAMENTAL FORMULATION

Lumped capacitance thermal balance: Internal thermal energy rate $mc_p\frac{dT}{dt}$ balances convective heat loss $\dot{Q}_{\text{loss}} = hA_s(T - T_A(t))$. Rate of cooling is $-\frac{dT}{dt}$.

PHASE 3: STEP-BY-STEP DERIVATION

Part (i): Identification of Variables and Parameters:

- **Independent variable:** Time t $[s]$ (or $[\text{min}]$, $[\text{h}]$).
- **Dependent variable:** Temperature of the hot chocolate $T = T(t)$ $[^\circ\text{C}]$ or $[\text{K}]$.
- **Parameters:**
  - Cooling constant: $k = \frac{hA_s}{mc_p} > 0$ $[t^{-1}]$.
  - Initial condition: $T(0) = T_0$ $[^\circ\text{C}]$.
  - Ambient temperature: $T_A(t)$ (specified external driving function; acts as a parameter if constant).

Part (ii): Formulation of Newton's Law of Cooling:

The verbal statement dictates that the cooling rate $-\frac{dT}{dt}$ is proportional to $(T(t) - T_A(t))$ with constant $k > 0$:

$$-\frac{dT}{dt} = k(T(t) - T_A(t))$$

Multiplying by -1 and rearranging to standard linear form:

$$\mathbf{\frac{dT}{dt} + kT(t) = kT_A(t)} \tag{1.4.1}$$

PHASE 4: FINAL RESULT & LINEARITY CHECK

BOXED RESULT

$$\frac{dT}{dt} = -k(T - T_A(t)) \iff \frac{dT}{dt} + kT = kT_A(t)$$

Linearity: In form $a_1(t)T' + a_0(t)T = b(t)$ with $a_1(t) = 1, a_0(t) = k, b(t) = kT_A(t)$. Strictly 1st-order linear ODE.

Forensic Time of Death Estimation (Algor Mortis)

Original statement · ProblemsCh1.pdf, p. 2

PHASE 1: PROBLEM STATEMENT & HYPOTHESES

Problem statement: A body is found in a cold room ($T_A = 5^\circ\text{C}$) at 3 pm, and its temperature then is 19°C. An hour later its temperature has dropped to 15°C. Use Newton's law of cooling to estimate the time of death, assuming that the living body temperature is 37°C and measuring time in hours.

PHASE 2: FUNDAMENTAL FORMULATION

Let $t = 0$ h correspond to 3:00 pm. Known data: $T_A = 5^\circ\text{C}$, $T(0) = 19^\circ\text{C}$, and $T(1) = 15^\circ\text{C}$ (at 4:00 pm). We seek the negative time $t_d < 0$ when $T(t_d) = 37^\circ\text{C}$.

PHASE 3: STEP-BY-STEP DERIVATION

Step 1: General Solution:

$$\frac{dT}{dt} = -k(T - 5) \implies \frac{d(T-5)}{T-5} = -k\,dt \implies T(t) - 5 = [T(0) - 5]e^{-kt}$$

With $T(0) = 19^\circ\text{C}$, the temperature difference is $19 - 5 = 14^\circ\text{C}$:

$$T(t) = 5 + 14e^{-kt} \tag{1.5.1}$$

Step 2: Determination of k:

At $t = 1$ h, $T(1) = 15^\circ\text{C}$:

$$15 = 5 + 14e^{-k} \implies 10 = 14e^{-k} \implies e^{-k} = \frac{10}{14} = \frac{5}{7}$$

Inverting both sides:

$$e^{k} = \frac{7}{5} = 1.4 \implies \mathbf{k = \ln(1.4) \approx 0.336472\ h^{-1}}$$

Step 3: Calculating Time of Death t_d:

Enforce the living core body temperature $T(t_d) = 37^{\circ}\text{C}$:

$$37 = 5 + 14e^{-kt_d} \implies 32 = 14e^{-kt_d} \implies e^{-kt_d} = \frac{32}{14} = \frac{16}{7}$$

Take the natural logarithm:

$$-kt_d = \ln\left(\frac{16}{7}\right) \implies t_d = -\frac{\ln(16/7)}{k} = -\frac{\ln(16/7)}{\ln(1.4)}$$

Evaluating numerically with $\ln(16/7) \approx 0.8266786$ and $\ln(1.4) \approx 0.3364722$:

$$t_d = -\frac{0.8266786}{0.3364722}\ \text{h} \approx \mathbf{-2.4569}\ \text{hours}$$

Step 4: Clock Time Conversion:

Multiply the fractional part by 60 minutes:

$$2.4569\ \text{hours} = 2\ \text{h} + (0.4569 \times 60)\ \text{min} = 2\ \text{h}\ 27.41\ \text{min} \approx 2\ \text{hours and}\ 27\ \text{minutes}$$

Subtracting 2 hours and 27 minutes from 3:00 pm (15:00):

$$15:00 - 2\ \text{h}\ 27\ \text{min} = \mathbf{12:33}\ \text{pm}$$

PHASE 4: FINAL RESULT & ANALYSIS

BOXED RESULT

$$t_d \approx -2.457 \text{ hours} \implies \text{Time of Death} \approx \mathbf{12} : \mathbf{33} \text{ pm}$$

Forensic Consistency: Algor mortis cooling is monotonic and deterministic under steady ambient conditions.

PROBLEM 1.6 | CSTR Mass Balance • Washout • Integrating Factor

Saline Mixing Tank Dynamics (CSTR)

Original statement · ProblemsCh1.pdf, p. 2

PHASE 1: PROBLEM STATEMENT & HYPOTHESES

Problem statement: A tank contains 100 liters of water with 1 kg of salt dissolved in it (uniformly). At time $t_0 = 0$, water without salt flows into the tank at a rate of 1 liter per minute, while the mixture flows out at the same rate.

(i) Write a differential equation for $x(t)$, which is the number of kg of salt per litre of water dissolved at time $t > 0$.
(ii) How much salt is dissolved in the tank as $t \to \infty$?
(iii) What happens if water with s kg of salt per liter flows into the tank at a given rate while the mixture (that is kept uniform) flows out at the same rate?

PHASE 2: FUNDAMENTAL FORMULATION

As in the official statement, $x(t)$ is the salt **concentration** (kg of salt per litre, units $\mathrm{kg/L}$). The total dissolved salt mass is $M(t) = V(t)\,x(t)$ (units kg). Species conservation of dissolved salt mass: $\frac{dM}{dt} = \dot{m}_{\text{in}} - \dot{m}_{\text{out}} = Q_{\text{in}} C_{\text{in}} - Q_{\text{out}}\,x(t)$. Since $Q_{\text{in}} = Q_{\text{out}} = Q$, $V(t) \equiv V_0 = 100$ L, so $M(t) = 100\,x(t)$ and the initial concentration is $x(0) = \frac{1\ \text{kg}}{100\ \text{L}} = 0.01\ \mathrm{kg/L}$.

PHASE 3: STEP-BY-STEP DERIVATION

Part (i): Pure Water Flushing ($C_{\text{in}} = 0$):

$$\dot{m}_{\text{in}} = 1 \times 0 = 0\ \mathrm{kg/min}, \quad \dot{m}_{\text{out}} = 1 \times x(t) = x(t)\ \mathrm{kg/min}$$

With $M = 100\,x$, the balance $\frac{dM}{dt} = \dot{m}_{\text{in}} - \dot{m}_{\text{out}}$ becomes $100\,\frac{dx}{dt} = -x(t)$, i.e.

$$\boxed{\frac{\mathbf{dx}}{\mathbf{dt}} = -\frac{\mathbf{x(t)}}{\mathbf{100}}, \quad x(0) = 0.01\ \mathrm{kg/L}} \qquad (1.6.1)$$

Part (ii): Analytical Solution and Limit as $t \to \infty$:

$$\int_{0.01}^{x(t)} \frac{d\xi}{\xi} = \int_0^t -\frac{1}{100} ds \implies \ln \frac{x(t)}{0.01} = -\frac{t}{100} \implies \mathbf{x(t) = 0.01\, e^{-t/100}\ \text{kg/L}}$$

The dissolved salt mass is $M(t) = 100\, x(t) = e^{-t/100}$ kg. Taking the limit as $t \to \infty$:

$$\lim_{t\to\infty} x(t) = \mathbf{0}\ \text{kg/L}, \qquad \lim_{t\to\infty} M(t) = \lim_{t\to\infty} e^{-t/100} = \mathbf{0}\ \text{kg}$$

Part (iii): Saline Inflow at Concentration s kg/L:

Now $\dot{m}_{\text{in}} = Q \cdot s = 1 \cdot s = s$ kg/min, so $100\, \frac{dx}{dt} = s - x(t)$, leading to the inhomogeneous linear ODE:

$$\frac{dx}{dt} + \frac{1}{100} x(t) = \frac{s}{100} \tag{1.6.2}$$

Multiplying by integrating factor $\mu(t) = \exp\left(\int \frac{1}{100} dt\right) = e^{t/100}$:

$$\frac{d}{dt}\left[x(t) e^{t/100}\right] = \frac{s}{100} e^{t/100}$$

Integrating both sides from 0 to t (Barrow's rule):

$$x(t) e^{t/100} - x(0) = \frac{s}{100} \int_0^t e^{\tau/100} d\tau = s\left[e^{t/100} - 1\right]$$

With $x(0) = 0.01$ kg/L:

$$x(t) e^{t/100} = 0.01 + s e^{t/100} - s = s e^{t/100} + (0.01 - s)$$

Dividing by $e^{t/100}$:

$$\mathbf{x(t) = s + (0.01 - s) e^{-t/100}\ \text{kg/L}} \tag{1.6.3}$$

The corresponding dissolved salt mass is $M(t) = 100\, x(t) = 100s + (1 - 100s) e^{-t/100}$ kg. As $t \to \infty$, $e^{-t/100} \to 0$:

$$\lim_{t\to\infty} x(t) = \mathbf{s}\ \text{kg/L}, \qquad \lim_{t\to\infty} M(t) = \mathbf{100s}\ \text{kg}$$

PHASE 4: FINAL RESULT & ANALYSIS

BOXED RESULT

$$\text{Washout: } x(t) = 0.01\, e^{-t/100}\ \text{kg/L},\ M(t) = e^{-t/100}\ \text{kg},\ \lim_{t\to\infty} M(t) = 0$$

$$\text{Saline Inflow: } x(t) = s + (0.01 - s)e^{-t/100}\ \text{kg/L},\ \lim_{t\to\infty} x(t) = s\ \text{kg/L},\ \lim_{t\to\infty} M(t) = 100s\ \text{kg}$$

Physical Interpretation: The asymptotic equilibrium concentration is the inlet concentration s kg/L, i.e. a dissolved mass $100s$ kg (tank volume 100 L times s kg/L).

PROBLEM 1.7

Newton II • Successive Integration • Kinematics

Free Fall Motion under Gravity

Original statement · ProblemsCh1.pdf, p. 2

PHASE 1: PROBLEM STATEMENT & HYPOTHESES

> **Problem statement:** Consider an object in free fall from an initial point $y(0) = y_0$ and with initial velocity 0, under the action of gravity and neglecting air friction. Write a differential equation for the position of the object $y(t)$ and solve it.

PHASE 2: FUNDAMENTAL FORMULATION

Coordinate y oriented vertically upward. Forces: $F_g = -mg\hat{j}$. Newton's second law: $m\frac{d^2y}{dt^2} = -mg \implies \frac{d^2y}{dt^2} = -g$. Initial data: $y(0) = y_0, y'(0) = 0$.

PHASE 3: STEP-BY-STEP DERIVATION

Stage 1: Velocity Integration:

$$\frac{d}{dt}\left(\frac{dy}{dt}\right) = -g \implies \int_0^t \frac{d}{ds}\left(\frac{dy}{ds}\right) ds = \int_0^t -g\, ds$$

$$\frac{dy}{dt}(t) - \frac{dy}{dt}(0) = -gt \implies v(t) = \frac{dy}{dt} = -gt \tag{1.7.1}$$

Stage 2: Position Integration:

$$\int_0^t \frac{dy}{ds}\, ds = \int_0^t -gs\, ds \implies y(t) - y(0) = \left[-\frac{1}{2}gs^2\right]_0^t$$

$$y(t) - y_0 = -\frac{1}{2}gt^2 \implies \mathbf{y(t) = y_0 - \frac{1}{2}gt^2} \tag{1.7.2}$$

PHASE 4: FINAL RESULT & ENERGY CONSERVATION

BOXED RESULT

$$y(t) = y_0 - \frac{1}{2}gt^2, \quad v(t) = -gt$$

Energy Check: $E = \frac{1}{2}mv^2 + mgy = \frac{1}{2}m(-gt)^2 + mg(y_0 - \frac{1}{2}gt^2) = mgy_0 = \text{constant}$.
Mechanical energy is strictly conserved.

PROBLEM 1.8

Newtonian Polar Frame • Mechanical Energy • Nonlinear ODE

Simple Pendulum Equation of Motion

Original statement · ProblemsCh1.pdf, p. 2

PHASE 1: PROBLEM STATEMENT & HYPOTHESES

> **Problem statement:** Consider a simple pendulum consisting of a bob of mass m at the end of a string of length l and negligible mass, and denote by $\theta(t)$ the angle with respect to the vertical as a function of time t. Neglecting friction, apply Newton's second law to write a differential equation for $\theta(t)$.

PHASE 2: FUNDAMENTAL FORMULATION

We present two independent methods: (1) Direct force projection in polar coordinates $(\hat{e}_r, \hat{e}_\theta)$ using Newton's second law; (2) Time differentiation of total mechanical energy $\frac{dE}{dt} = 0$.

PHASE 3: STEP-BY-STEP DERIVATION

Method 1: Polar Coordinate Force Balance:

Bob position: $r = l\hat{e}_r$. Taking time derivatives with $\dot{\hat{e}}_r = \dot{\theta}\hat{e}_\theta$ and $\dot{\hat{e}}_\theta = -\dot{\theta}\hat{e}_r$:

$$v = l\dot{\theta}\hat{e}_\theta$$

$$a = \frac{dv}{dt} = -l\dot{\theta}^2\hat{e}_r + l\ddot{\theta}\hat{e}_\theta$$

The tangential acceleration is $a_\theta = l\ddot{\theta}$.

Acting forces: String tension $T = -T\hat{e}_r$ acts purely radially. Gravity resolves as:

$$F_g = mg\cos\theta\,\hat{e}_r - mg\sin\theta\,\hat{e}_\theta$$

Projecting Newton's second law along the tangential unit vector $\hat{e}_\theta$:

$$\sum F_\theta = ma_\theta \implies -mg\sin\theta = m(l\ddot{\theta})$$

Dividing by $ml \neq 0$ and bringing all terms to the left-hand side:

$$\ddot{\theta} + \frac{\mathbf{g}}{\mathbf{l}}\sin\theta = \mathbf{0} \tag{1.8.1}$$

Method 2: Energy Conservation Check:

$$E = E_k + E_p = \frac{1}{2}mv^2 + mgl(1-\cos\theta) = \frac{1}{2}ml^2\dot{\theta}^2 + mgl(1-\cos\theta)$$

Differentiating with respect to time t ($\frac{dE}{dt} = 0$):

$$\frac{dE}{dt} = ml^2\dot{\theta}\ddot{\theta} + mgl\sin\theta\,\dot{\theta} = \dot{\theta}\left[ml^2\ddot{\theta} + mgl\sin\theta\right] = 0$$

For non-trivial motion ($\dot{\theta} \not\equiv 0$), dividing by ml^2 yields identically: $\ddot{\theta} + \frac{\mathbf{g}}{\mathbf{l}}\sin\theta = \mathbf{0}$.

PHASE 4: FINAL RESULT & ANALYSIS

BOXED RESULT

$$\ddot{\theta}(t) + \frac{g}{l}\sin\theta(t) = 0$$

Classification: 2nd-order, autonomous, **nonlinear** ODE due to the transcendental term $\sin\theta$. For small amplitudes ($\theta \ll 1$, $\sin\theta \approx \theta$), it linearizes to the harmonic oscillator $\ddot{\theta} + \omega_0^2\theta = 0$ with natural frequency $\omega_0 = \quad g/l$.

PROBLEM 1.9

Partial Fractions • Carrying Capacity • Malthusian Limit

Logistic Population Growth Model (Verhulst)

Original statement · ProblemsCh1.pdf, p. 2

PHASE 1: PROBLEM STATEMENT & HYPOTHESES

Problem statement: A refinement over Malthus' population growth model discussed in Chapter 1 is provided by the logistic equation:

$$\frac{dp}{dt} = kp\left(1 - \frac{p}{M}\right)$$

where $p(t)$ is the population size at time t, k is its growth rate (difference between birth rate and death rate), and M is the maximum sustainable population (carrying capacity).

Solve the equation for a generic initial condition $p(t_0) = p_0$ and discuss the evolution of the population for $t \to \infty$, comparing with Malthus' ($M \to \infty$) prediction.

PHASE 2: FUNDAMENTAL FORMULATION

First-order nonlinear autonomous ODE. Separate variables by moving all terms in p to the LHS, decompose into partial fractions, and integrate from (t_0, p_0) to (t, p).

PHASE 3: STEP-BY-STEP DERIVATION

Step 1: Separation of Variables:

$$\frac{dp}{dt} = \frac{k}{M}p(M-p) \implies \frac{dp}{p(M-p)} = \frac{k}{M}\,dt \tag{1.9.1}$$

Step 2: Partial Fraction Decomposition:

$$\frac{1}{p(M-p)} = \frac{A}{p} + \frac{B}{M-p} \implies 1 = A(M-p) + Bp$$

At $p = 0 \implies A = \frac{1}{M}$; at $p = M \implies B = \frac{1}{M}$. Substituting back:

$$\frac{1}{M}\left(\frac{1}{p} + \frac{1}{M-p}\right)dp = \frac{k}{M}\,dt \implies \left(\frac{1}{p} + \frac{1}{M-p}\right)dp = k\,dt \tag{1.9.2}$$

Step 3: Definite Integration:

$$\int_{p_0}^{p}\left(\frac{1}{u} + \frac{1}{M-u}\right)du = \int_{t_0}^{t} k\,ds$$

Recall $\int \frac{1}{M-u}du = -\ln|M-u|$:

$$\left[\ln|u| - \ln|M-u|\right]_{p_0}^{p} = k(t-t_0) \implies \left[\ln\left|\frac{u}{M-u}\right|\right]_{p_0}^{p} = k(t-t_0)$$

$$\ln\left|\frac{p(M-p_0)}{(M-p)p_0}\right| = k(t-t_0)$$

Step 4: Exponentiation and Isolation of $p(t)$:

$$\frac{p(M-p_0)}{(M-p)p_0} = e^{k(t-t_0)} \implies p(M-p_0) = p_0(M-p)e^{k(t-t_0)}$$

$$p(M-p_0) + pp_0e^{k(t-t_0)} = Mp_0e^{k(t-t_0)} \implies p\left[(M-p_0) + p_0e^{k(t-t_0)}\right] = Mp_0e^{k(t-t_0)}$$

Dividing numerator and denominator by $e^{k(t-t_0)}$:

$$\mathbf{p(t) = \frac{Mp_0}{p_0 + (M-p_0)e^{-k(t-t_0)}}} \tag{1.9.3}$$

Step 5: Asymptotic Limits:

- As $t \to \infty$: $\lim_{t\to\infty} e^{-k(t-t_0)} = 0 \implies \lim_{t\to\infty} p(t) = \frac{Mp_0}{p_0+0} = \mathbf{M}$.

- As $M \to \infty$ (Recovery of Malthus): Dividing numerator and denominator of (1.9.3) by M:

$$p(t) = \frac{p_0}{\frac{p_0}{M} + \left(1 - \frac{p_0}{M}\right) e^{-k(t-t_0)}} \implies \lim_{M\to\infty} p(t) = \frac{p_0}{0 + (1-0)e^{-k(t-t_0)}} = \mathbf{p_0 e^{k(t-t_0)}}$$

PHASE 4: FINAL RESULT & ANALYSIS

BOXED RESULT

$$p(t) = \frac{Mp_0}{p_0 + (M - p_0)e^{-k(t-t_0)}}$$

$$\lim_{t\to\infty} p(t) = M, \quad \lim_{M\to\infty} p(t) = p_0 e^{k(t-t_0)}$$

Physical Interpretation: The logistic curve exhibits a sigmoidal shape when $p_0 < M/2$, passing through an inflection point of maximum growth rate at $p = M/2$.

PROBLEM 1.10

Navier-Stokes Reduction • Centerline Regularity • No-Slip Wall

Laminar Viscous Poiseuille Flow from Navier-Stokes

Original statement · ProblemsCh1.pdf, p. 2

PHASE 1: PROBLEM STATEMENT & HYPOTHESES

Problem statement: In general, it is not possible to find explicit solutions of the Navier-Stokes equations that govern fluid flow. However, in certain cases the equations simplify very strongly. Suppose that a fluid is flowing down a pipe that has a circular cross-section of radius a. Assuming that the velocity V of the fluid depends only on its distance from the centre of the pipe, the equation satisfied by V is:

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{dV}{dr}\right) = -P$$

where $P > 0$.

Multiply by r and integrate once to show that:

$$\frac{dV}{dr} = -\frac{Pr}{2} + \frac{c}{r}$$

where c is an arbitrary constant.

Integrate again to find an expression for the velocity, and then use the facts that:
(i) the velocity should be finite at all points in the pipe, and
(ii) fluids 'stick' to boundaries (which means that $V(a) = 0$),
to show that:

$$V(r) = \frac{P}{4}\left(a^2 - r^2\right)$$

(Poiseuille flow).

PHASE 2: FUNDAMENTAL FORMULATION

This is a linear second-order Boundary Value Problem (BVP) in radial coordinate $r \in [0, a]$. The parameter $P \equiv -\frac{1}{\mu}\frac{dp}{dz} > 0$ represents the pressure driving force per unit dynamic viscosity.

PHASE 3: STEP-BY-STEP DERIVATION

Step 1: First Integration (Velocity Gradient):

Multiply the differential equation by r:

$$\frac{d}{dr}\left(r\frac{dV}{dr}\right) = -Pr$$

Integrate directly with respect to r:

$$\int \frac{d}{dr}\left(r\frac{dV}{dr}\right) dr = \int -Pr\, dr \implies r\frac{dV}{dr} = -P\frac{r^2}{2} + c$$

Dividing by r ($r > 0$):

$$\mathbf{\frac{dV}{dr} = -\frac{Pr}{2} + \frac{c}{r}} \qquad \boxed{(1.10.1)}$$

which rigorously proves the required intermediate formula.

Step 2: Second Integration (Velocity Field), keeping c:

Integrate (1.10.1) with respect to r on $r > 0$, using $\int r\, dr = \frac{r^2}{2}$ and $\int \frac{dr}{r} = \ln r$:

$$V(r) = \int \left(-\frac{Pr}{2} + \frac{c}{r}\right) dr = -\frac{P}{2}\cdot\frac{r^2}{2} + c\ln r + C_2 = -\frac{Pr^2}{4} + c\ln r + C_2 \qquad \boxed{(1.10.2)}$$

where C_2 is the second constant of integration.

Step 3: Condition (i), Finite Velocity at the Centerline ($r = 0$):

The velocity must be finite at every point of the pipe, including $r = 0$. Since $\lim_{r\to 0^+} r^2 = 0$ but $\lim_{r\to 0^+} \ln r = -\infty$, the term $c\ln r$ makes $V(r) \to \mp\infty$ for any $c \neq 0$ (and $\frac{dV}{dr} = \frac{c}{r}$ diverges as well, so the shear stress $\tau_{rz} = \mu\frac{dV}{dr}$ would be unbounded). Finiteness therefore requires:

$$\mathbf{c = 0} \implies V(r) = -\frac{Pr^2}{4} + C_2 \qquad \boxed{(1.10.3)}$$

Step 4: Enforcement of No-Slip Boundary Condition ($r = a$):

Condition (ii) states that viscous fluids adhere to solid surfaces without slip: $V(a) = 0$. Imposing this on equation (1.10.3):

$$V(a) = -\frac{Pa^2}{4} + C_2 = 0 \implies \mathbf{C_2 = \frac{Pa^2}{4}}$$

Substituting C_2 back into (1.10.3):

$$V(r) = \frac{Pa^2}{4} - \frac{Pr^2}{4} = \mathbf{\frac{P}{4}\left(a^2 - r^2\right)} \tag{1.10.4}$$

PHASE 4: FINAL RESULT & AEROSPACE QUANTITIES

BOXED RESULT

$$\frac{dV}{dr} = -\frac{Pr}{2} + \frac{c}{r} \implies V(r) = \frac{P}{4}\left(a^2 - r^2\right)$$

Engineering Metrics:

- Centerline peak velocity: $V_{\max} = V(0) = \frac{Pa^2}{4}$.
- Volumetric Flow Rate (Hagen-Poiseuille Law), integrating over the annular elements $dA = 2\pi r\, dr$ with Barrow's rule:

$$Q = \int_0^a \frac{P}{4}\left(a^2 - r^2\right) 2\pi r\, dr = \frac{\pi P}{2}\int_0^a \left(a^2 r - r^3\right) dr = \frac{\pi P}{2}\left[\frac{a^2 r^2}{2} - \frac{r^4}{4}\right]_0^a = \frac{\pi P}{2}\left(\frac{a^4}{2} - \frac{a^4}{4}\right) = \frac{\pi P a^4}{8}$$

  With $P = -\frac{1}{\mu}\frac{dp}{dz}$ (see Phase 2) this gives $Q = \frac{\pi a^4}{8\mu}\left(-\frac{dp}{dz}\right)$.
- Wall shear stress: $\tau_w = -\mu\left.\frac{dV}{dr}\right|_{r=a} = \frac{\mu P a}{2} = \frac{a}{2}\left(-\frac{dp}{dz}\right)$.

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